

Lab Assignment Questions

Q1. Analysis of Image Enhancement Techniques

Apply for given **grayscale images** representing different conditions.

Tasks:

a) Apply the following point processing techniques:

- Image Negative
- Log Transformation
- Gamma Correction (use at least two different gamma values)
- Contrast Stretching

b) For each technique:

- Display the original and processed images
- Plot the corresponding histograms

c) Analyze the results and explain:

- How each technique affects image intensity distribution
- Which method is most suitable for each type of image and why

Q2. Histogram Equalization: Theory and Implementation

Tasks:

a) Starting from the definition of histogram, derive the histogram equalization transformation using the cumulative distribution function (CDF).

b) Consider the given small 8-bit grayscale image.

- Compute the histogram
- Calculate probability distribution
- Determine the CDF
- Obtain the equalized intensity values manually

c) Implement histogram equalization using:

- A built-in function
- A manual algorithm (without using library functions)

d) Compare and discuss the results in terms of contrast improvement.

8×8 Grayscale Pixel Values

52	55	61	59	79	61	61
62	59	55	104	94	89	71
63	65	66	113	144	104	62
64	70	70	126	154	109	63
67	73	68	106	122	88	68
68	79	60	70	77	68	75
89	65	64	58	55	61	83
70	87	69	65	73	78	90

Q3. Modeling and Simulation of a Unicycle Robot

Tasks:

- a) Derive the kinematic equations of a unicycle robot and explain the meaning of each variable.
- b) Explain the concept of the non-holonomic constraint and its physical significance.
- c) Simulate the robot motion under the following conditions:
 - Straight-line motion
 - Pure rotational motion
 - Circular motion
- d) Plot the trajectory (X–Y plane) for each case and interpret the results.

Q4. Design and Performance Analysis of PID Controller

Tasks:

- a) Define proportional, integral, and derivative control actions and explain their individual roles.
- b) Develop a PID controller for a unicycle robot to move from an initial position to a desired target position.
- c) Implement and simulate the following controllers:
 - P controller
 - PI controller
 - PD controller
 - PID controller
- d) For each case, plot:
 - Robot trajectory
 - Error versus time
- e) Compare the performance in terms of:
 - Convergence speed
 - Overshoot
 - Steady-state error
 - Stability

Q5. Linear Quadratic Regulator (LQR) Design

Tasks:

- a) Explain the concept of LQR and formulate the cost function used in optimal control.
- b) For a given linear system, derive the optimal control law and explain the role of matrices (Q) and (R).
- c) Analyze how varying (Q) and (R) affects system behavior (aggressiveness vs smoothness).
- d) Compare LQR with PID control in terms of design approach, performance, and limitations.

Q6. LQR-Based Control of a Unicycle System

Tasks:

- a) Convert the nonlinear unicycle model into an error-state representation.
- b) Linearize the system around an operating point.
- c) Design an LQR controller by selecting appropriate (Q) and (R) matrices.
- d) Simulate the robot motion from an initial position to a desired goal position.
- e) Compare the results with PID control in terms of:
 - Path smoothness
 - Control effort
 - Accuracy of convergence